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Design Document

Capstone Deliverable 1: architecture, agents, checkpoints and governance

AI-driven Project Management System for the Harbourline Community Hub Refurbishment

Course: MA6094 Agentic AI for Project Management
Capstone Project Group 5

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Abstract

This document describes the design of an AI-driven project management system for the Harbourline Community Hub Refurbishment, as built in the latest code. Six agent modules and a governance layer plan, track and report the refurbishment behind one local web user interface. The modules are Planning, Tool Integration, Memory & Retrieval, Risk & What-if, Reporting with its Critic, and Change. The system writes the approved plan into Jira Cloud or into a local mock board that implements the same contract. The core design principle is that *the LLM drafts and deterministic code decides*: report numbers and dates come from verified facts, the LLM drafts structured content that must cite its sources, validators and a Critic check every draft, and a curated baseline is used when no model is configured or validation fails. Gemini is configured first and DeepSeek as backup. Seven human checkpoints, HC1 to HC7, sit between agent output and anything a stakeholder or the PM tool sees; four of them gate code paths, while HC1, HC4 and HC7 only record. The document covers the architecture, the agent cards, the checkpoint map, the data contract with the PM tool, change handling, governance and free-tier compliance, and the design trade-offs. It also states the known limitations, among them the passwordless demo identities and the project-wide Reset demo added on 11 Nov 2026, which the team has accepted, and the evidence that is kept outside the repository.

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1 Purpose, scope and design principles

1.1 Document conventions

Team. Group 5 has no team name. Its members are Tian Yujia, G2510505D, team lead, responsible for the Design Document and the demo system; Qu Chengzhi, G2510533D, Evaluation Report; Huang Hao, G2510626K, Governance Log; Zhang Yinan, G2505943K, demo video; Liu Beiqi, G2510916G, slides; and Zhao Lingfeng, G2510595H. The agent modules have no individual owners; the team lead is responsible for all of them.

Repository. The public repository is `github.com/FrundlesTian/harbourline-pm`, branch `main`, latest commit.

Evidence convention. `path:lines` refers to the repository; `BRIEF-p3-5`, `SA-Q1`, `EDC-2.3`, `NRF-4.2` and similar IDs are the system's own source IDs, listed in Appendix E. Requirement references such as `L137` are line numbers in `data_pack/00_Brief_UPDATED.txt`.

1.2 Purpose and scope

Six agent modules and a governance layer plan, track and report the Harbourline refurbishment behind one local web UI, `server.py`, and write the approved plan into Jira Cloud. Requirements come from the UPDATED brief, L137–L201, only.

1.3 Design principles

Core principle — the LLM drafts, deterministic code decides. It is implemented as four rules:

1. **Facts are verified, not generated.** Report numbers and dates come from facts F01–F78. Each fact is rejected at load unless its exact quote is in its source chunk, see `app/facts_catalog.py:89-108`. Conflict detectors CF01–CF12 compute from these facts in `app/planning.py:219-281`.
2. **The LLM drafts structured content and must cite.** By default the plan, risk register, report bodies and change analysis are generated as JSON carrying source and fact IDs, see `app/planning.py:327-487`, `app/risk.py:173-235`, `app/reporting.py:161-238` and `app/change.py:128-191`. The grounding rule is in `app/gen.py:73-76`.
3. **Validators and a Critic check every draft.** Unknown IDs are dropped, WBS must trace to an in-scope line, and a date survives only if a cited source states it, see `app/planning.py:423-432`. The Critic traces every number, date and person in a report to what the line cites, see `app/critic.py:82-120`.
4. **A curated baseline is the fallback** when no model is configured or validation fails. The reason is recorded by `app/gen.py:61-66` and shown on the Overview page. The committed baseline evaluation passes 34 of 34 cases in this mode, see `eval/results/latest.md`; the LLM mode also passes 34 of 34, as reported in the Evaluation Report.

Human checkpoints HC1–HC7 sit between agent output and anything a stakeholder or the PM tool sees; Chapter 4 maps them. Open questions and assumptions are kept in the assumption register in Appendix A, which the Planning Agent seeds automatically in `app/planning.py:513`.

2 Architecture

Figure 2-1 shows the agents, the stores they read and write, the human checkpoints and the PM tool. It is redrawn from the full Mermaid diagram in `docs/design/architecture.mmd`.

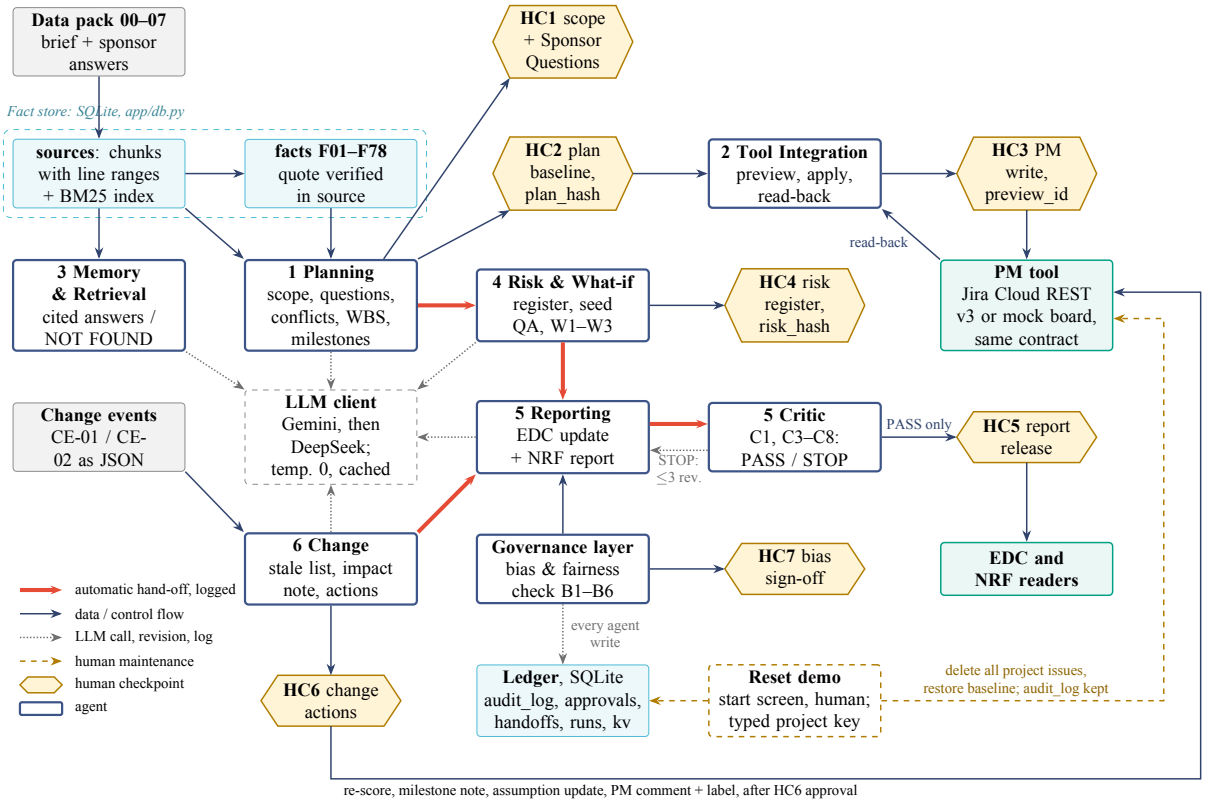


Figure 2-1 System architecture. Thick orange arrows are automatic hand-offs: Planning → Risk carries WBS + milestones, Risk → Reporting the register + what-ifs, Reporting → Critic the draft + claims, and Change → Reporting the refreshed stale reports. Each hand-off is written to the `handoffs` table by `audit.handoff`, see `app/audit.py:52–57`. Yellow hexagons are human checkpoints; dotted grey arrows are LLM calls, the Critic's revision loop and log writes; the dashed Reset demo path is a human maintenance action outside the agents, described in Chapter 5.

2.1 Stack as built

The system uses the Python 3 standard library only: `ThreadingHTTPServer` in `server.py:482–499`, a vanilla HTML/JS front end in `static/`, and SQLite for every store including the audit log, see `app/db.py`. The LLM client calls OpenAI-compatible chat endpoints and tries providers in order: **Gemini via Google AI Studio first, DeepSeek as backup**, see `app/config.py:27–59` and `app/llm.py:66–115`. A failed call is logged and retried on the next provider, see `app/llm.py:72–80`. The PM tool is Jira Cloud REST v3, with a local mock board that implements the same contract in `app/pmtool.py:71–135`.

Since 11 Nov 2026 there is **no password**, by design for the demo. The start screen lists the active accounts and the user picks one, see `app/auth.py:129–148` and `GET /api/identities`. The chosen account's role is then enforced server-side on every call, see `app/auth.py:184–202`

and `server.py:352–378`. The start screen also offers **Reset demo**, described in Chapter 5. There is no Streamlit UI and no JSONL log; the UI runs locally at `http://127.0.0.1:8000` and opens in English, as set in `app/config.py` and `static/i18n.js`.

2.2 One full run

One full run is started by `POST /api/run` and implemented in `app/pipeline.py:21–33`. It proceeds as follows:

1. `init` chunks the pack into 415 line-addressed chunks, builds the BM25 index (Robertson & Zaragoza, 2009), verifies the facts and restores human decisions, see `app/pipeline.py:10–18`.
2. **Planning** generates scope, Sponsor Questions, WBS and milestones, runs the conflict detectors, filters untraceable WBS elements and hands the WBS to Risk, see `app/planning.py:490–561`.
3. **Risk** generates the register from the seed register, pack evidence and WBS, validates it, computes what-ifs W1–W3 and hands off to Reporting, see `app/risk.py:238–279`.
4. **Governance** recomputes the bias check in `app/governance.py:43–107`.
5. **Reporting** drafts the EDC update and NRF report; the Critic checks each draft, and STOP findings go back to the model up to three times, see `app/reporting.py:221–238`.
6. Tokens, cost and seconds are recorded per run in `app/audit.py:89–116`.

Then a person approves HC2, Tool integration shows a read-only diff, a person approves HC3 for that diff, and `apply` writes and reads back, see `app/pmtool.py:482–592` and Figure 5-1. Temperature 0.0 plus a response cache keyed on model chain + prompt, implemented in `app/llm.py:34–63`, make a re-run on the same database reproduce the plan, register and board, so HC2/HC4 hashes stay valid.

3 Agent cards

Role and Context are fixed in `app/agents.py:9–91`; Appendix B gives the full text. **Model for every LLM call:** Gemini `gemini-2.5-flash` by default, set by the team in `.env` and read in `app/config.py:33`, with DeepSeek `deepseek-chat` as backup, see `app/config.py:42`; temperature 0.0, see `app/llm.py:66`. The Gemini path is tested only against a local stub; every LLM run the team made used DeepSeek, whose use the course instructor has accepted according to a team statement of 14 Nov 2026.

Tables 3-1–3-8 give one card per agent module and one for the governance layer.

3.1 Planning Agent

Table 3-1 Agent card: 1 Planning Agent

Role prompt	`app/agents.py:10–21`; generation prompts `app/planning.py:327–474`
Inputs	Brief chunks, sponsor answers, NRF, EDC, DEL, LL chunks; fact store
Outputs	Scope lines SC/OS; Sponsor Questions GQnn with status, answer and assumption; conflict register CF01–CF12; WBS to L3 with dictionary; flagged elements; milestones MSnn with dependencies
Tools/APIs	LLM returning JSON; validators; deterministic conflict detectors
Stop condition	Generated plan rejected if <8 scope lines, <10 Sponsor Questions, L1<4 or L3<25, or <8 milestones; `app/planning.py:373–376`, `416–417`, `472–473`
Failure behaviour	If the answer stops after the scope, the Sponsor Question register is requested once more on its own, `app/planning.py:344–347`. Otherwise falls back to baseline tables; the reason is shown on Overview and in audit as `output generated (baseline)`, outcome `fallback`, `app/gen.py:61–66`. The optional extra-gaps call is cached like every other call, `app/planning.py:296`
Evidence	Eval T01–T05 in `eval/results/latest.md`; LLM-mode log `docs/evaluation/evidence/eval_llm_deepseek_2026-11-14_final.log`

3.2 Tool Integration Agent

Table 3-2 Agent card: 2 Tool Integration Agent

Role prompt	`app/agents.py:22–30` — "Deterministic agent - no LLM"
Inputs	HC2-approved WBS + milestones; HC6-approved change actions
Outputs	Preview listing create/update/unchanged with a `preview_id`; PM issues; read-back report
Tools/APIs	Jira Cloud REST v3, see Chapter 5, or the mock board, both behind `Guarded`, `app/pmtool.py:343–390`
Stop condition	No write without HC2 for the current plan hash and HC3 for the preview; `app/pmtool.py:485–486`, `528–531`
Failure behaviour	Refusal returned and logged `pm sync refused` with outcome blocked; Jira errors 401/403/404 surface with a hint and the token redacted, `app/pmtool.py:151–172`
Evidence	Eval T21–T23, T30 on the mock board

3.3 Memory & Retrieval Agent

Table 3-3 Agent card: 3 Memory & Retrieval Agent

Role prompt	`app/agents.py:31–40`
Inputs	Question; BM25 top hits; decision log DL-nnn
Outputs	Answer with document + line citations, or NOT FOUND with reason; conflict notes; `qa_log` row
Tools/APIs	BM25 (Robertson & Zaragoza, 2009), `app/sources.py:294–332`; optional LLM synthesis
Stop condition	Deterministic sufficiency test: top score <2.5, key-term coverage $<70\%$, or no value where one is asked $\rightarrow$ NOT FOUND; `app/memory.py:76–87`
Failure behaviour	An LLM answer with an unknown citation or untraceable number is rejected and logged; the extractive answer is used instead, `app/memory.py:105–122`
Evidence	Eval T06–T15

3.4 Risk & What-if Agent

Table 3-4 Agent card: 4 Risk & What-if Agent

Role prompt	`app/agents.py:41–51`; generation prompt `app/risk.py:173–202`
Inputs	Seed register RISK-Rnn, pack chunks, WBS hand-off, conflicts
Outputs	Register with L×I, owner, trigger, mitigation, WBS links and sources; seed QA findings; what-ifs W1–W3, plus W4 for CE-02
Tools/APIs	LLM returning JSON; validators; deterministic seed QA and what-if arithmetic, `app/risk.py:110–126`, `301–381`
Stop condition	A risk is dropped if L or I is outside 1–5, it has no valid WBS link, no known source, no owner or trigger, or if a number or date in its description or trigger is not stated by its cited sources after one revision round, `app/risk.py:129–170`. The register is rejected if it has <12 valid risks, `app/risk.py:214–227`
Failure behaviour	Baseline register used; reason logged. A missing vendor-change risk is added by the validator, `app/risk.py:228–234`
Evidence	Eval T16, T17

3.5 Reporting Agent

Table 3-5 Agent card: 5 Reporting Agent

Role prompt	`app/agents.py:52–62`; report prompt `app/reporting.py:161–194`
Inputs	One fact sheet of facts with quotes, conflicts, change events, NRF/DEL chunks; top-5 risks inserted from the register, `app/reporting.py:197–218`
Outputs	EDC exception update and NRF monthly report, each with a claim list of text, cites and calc
Tools/APIs	LLM returning JSON; Critic; line templates, `app/reporting.py:67–125`
Stop condition	Draft only — cannot release; hands to Critic automatically, `app/reporting.py:285–287`
Failure behaviour	Still STOPPED after 3 revisions → template draft used and logged `generated draft rejected – template fallback`, `app/reporting.py:249–255`
Evidence	Eval T18, T25

3.6 Critic Agent

Table 3-6 Agent card: 5 Critic Agent

Role prompt	`app/agents.py:63–71` — "Deterministic agent - no LLM"
Inputs	Draft sections, claims, rendered content
Outputs	Findings with rule, severity and detail; report status `critic_passed` or `critic_stopped`
Tools/APIs	Number/date/person tracer with a safe formula evaluator, `app/critic.py:62–120`
Blocker rule	Any finding of severity STOP blocks release: C1 untraceable number/date/person or uncited claim; C3 missing required section; C4 top five ≠ register; C5 superseded fact or unreflected change event; C6 banned status phrase such as "on track" or "on programme" unless attributed to KHB; C7 conflict line with <2 sources; C8 NRF report without NOT FOUND; `app/critic.py:123–169`
Max revision loops	**3** for generated report drafts, `MAX_REVISIONS = 3` in `app/reporting.py:151`; **1** for generated change cost lines and for generated risks, after which untraceable lines or risks are dropped, `app/change.py:165–178`, `app/risk.py:137–163`; **0** for template and planted-fault drafts
Failure behaviour	A STOPPED report is stored with its findings; HC5 release is refused, `app/reporting.py:291–298`
Evidence	Eval T19 with 21/21 planted faults, T20, T28

3.7 Change Agent

Table 3-7 Agent card: 6 Change Agent

Role prompt	`app/agents.py:72–81`; analysis prompt `app/change.py:130–158`
Inputs	Event JSON `change_events/CE–01.json`, `CE–02.json`; tags of every stored output
Outputs	Stale list; Change Impact Note with cost/time, sponsor decisions and UNRESOLVED items; auto-refreshed what-ifs and reports; pending HC6 actions
Tools/APIs	Deterministic tag matching + milestone dependency walk, `app/change.py:38–58`, `248–282`; LLM analysis; Critic tracer
Stop condition	Never edits baseline, closes or deletes; analysis rejected if <2 traced cost lines or <2 sourced decisions, `app/change.py:208–221`
Failure behaviour	Baseline CE-01/CE-02 analysis used, `app/change.py:65–125`; rejected cost lines logged `reject generated cost line`. If affected risks exist but no valid re-score is proposed, the agent is asked once for re-scores only, `app/change.py:179–188`
Evidence	Eval T24–T28

3.8 Governance layer

Table 3-8 Card: governance layer

Role prompt	`app/agents.py:82–90` — deterministic
Inputs	Every agent write and approval; `05_Resident_consultation_responses.csv`
Outputs	`audit_log`, `approvals`, CSV exports; bias report B1–B6 with actions
Tools/APIs	SQLite; theme lexicon en/zh/ms/ta, `app/governance.py:9–41`
Stop condition	Bias check status "awaiting HC7" until signed off, `app/governance.py:100–112`
Failure behaviour	An unknown checkpoint, unnamed approver or invalid decision raises an error, `app/audit.py:60–66`
Evidence	Eval T20, T23, T29, T30

4 Human checkpoint map

Checkpoints are defined in `app/audit.py:16–31`. The server records every approval under the selected session account, never a typed name, see `server.py:118–143`, and checks that account's role first in `app/auth.py:184–202`. Approvals go to the `approvals` table with approver, role, account, decision, comment and timestamp; the table is defined in `app/db.py:58–60` with migrations at `:95–98`. Each approval is mirrored in `audit_log`. **The account is chosen without a password**, so an approval shows which account approved, not which person; this is limitation L7 in Section 8.2. The start-screen **Reset demo** deletes the `approvals` table together with the other demo data; the mirrored rows in `audit_log` remain, as Chapter 7 explains.

Table 4-1 lists the seven checkpoints, what each is bound to, and what happens if it is rejected or skipped. Keeping accountable decisions with named roles at defined points follows the governance practice in the PMBOK Guide (Project Management Institute, 2021).

Table 4-1 Human checkpoint map

ID	What is approved, bound to	Shown to the approver	Approver RBAC role and account	On reject	If skipped, what is stale or unprotected
HC1	Scope + Sponsor Questions register, `gaps_hash`	Planning page: scope, questions, answers, assumptions, conflicts	`project_manager`, Project Manager account	Rejection recorded; plan stays in draft	**Record only:** no code path is blocked by HC1. Questions reach the sponsor unreviewed.
HC2	Plan baseline, `plan_hash` of WBS + milestones	WBS, dictionary, milestones, dependency issues	`project_manager`, Project Manager account	`preview` refuses to run	**Enforced:** no PM sync is possible, `app/pmtool.py:485–486`; any plan change needs re-approval
HC3	One sync diff, `preview_id`	Per-item create / update / unchanged list	`project_manager`, Project Manager account	`apply` refuses, `app/pmtool.py:528–531`	**Enforced:** no write to Jira. Limitation: Section 8.2, L2
HC4	Risk register version, `risk_hash`	Register, seed QA, what-ifs	`group_leader`, Group Leader account	Rejection recorded	**Record only:** reports still draw top-five risks from an unapproved register
HC5	Release of one report	Report, claims, Critic findings	`group_leader`, Group Leader account	Report stays a draft	**Enforced:** only `critic_passed` drafts can be released, `app/reporting.py:291–301`
HC6	One change action: re-score, PM comment + label, milestone note or assumption update	Change Impact Note and the action payload	`project_manager`, Project Manager account	Action marked rejected; nothing applied	**Enforced:** actions apply only in `approve_action`, `app/change.py:388–431`; risks stay at old scores

Table 4-1 continued

ID	What is approved, bound to	Shown to the approver	Approver RBAC role and account	On reject	If skipped, what is stale or unprotected
HC7	Bias & fairness check BIAS-1	Findings B1–B6, re-weighted ranks, translations	`supervisor`, Supervisor account	Status "rejected"	**Record only:** status shown; no code consumes the consultation summary

The role strings in `app/audit.py`, such as "Team PM lead", are older labels; the RBAC role is what is checked.

4.1 Not automated, by design

- **Moving items to Done, closing, resolving or deleting.** The brief makes these human acts in L247–L248. `Guarded` blocks them and logs `BLOCKED`, see `app/pmtool.py:378–390`; the forbidden fields are `status`, `resolution` and `resolutiondate`, listed in `app/pmtool.py:29`. Read-back reports items a human moved to Done, see `app/pmtool.py:586`.
- **Sponsor decisions**, such as closure priority in `SA–Q3`, contingency status in `EDC–1.2` or rooftop deferral in `EDC–6.3`. These appear as decisions with who and when, or as NOT FOUND, never as choices made by the agent, see `app/change.py:188–190`.
- **Report release at HC5 and risk re-scoring at HC6.** Agents propose; a person accepts, because the reports go to external bodies and scores drive the top-five list.

5 Data contract with the PM tool

Table 5-1 maps each plan element to the PM field it is written to; line references are to `app/pmtool.py`.

Table 5-1 Mapping of plan elements to PM fields, with rules in app/pmtool.py

Plan element	PM field	Rule in `app/pmtool.py`
WBS L1 / L2 / L3	Issue type Epic / Task / Subtask	`:41–50`; Jira types resolved by hierarchy level, so localised type names work, `:245–266`
Milestone	Task, summary `[MSnn] Milestone: <name>`	`:52–61`
WBS parent	`parent` = parent's issue key	via ext-key map, parents written first, `:539–556`
Dependency	"Blocks" issue link	idempotent; direction learned once and corrected, `:294–317`
Labels	`ai-generated`, ext key, `wbs` or `milestone`	create refused without `ai-generated`, `:356–358`
Due date	`duedate` only for a full ISO date	otherwise the description says the date is NOT FOUND, `:53`, `57`
Description	WBS dictionary or milestone note + source citations	`:48`, `57`

5.1 External key and idempotency

Each item has a stable label key such as `ext-HCH-WBS-1-2-3`, built in `:37–38`. Sync has three steps. First, each item is looked up by ext key with one search plus a bulk read of keys written before, which covers Jira's index lag, see `:452–479`. Second, a SHA-256 content hash is compared with `pm_sync_map` and the board copy, see `:65–67` and `:515–518`. Third, the item is created, updated or left unchanged, see `:543–554`. `change-*` labels from approved HC6 actions survive re-syncs, see `:510–512`. In eval T22 the second sync created 0, updated 0 and left 92 unchanged.

5.2 Approval bound to the diff

`preview` hashes the list of ext key, action and hash into `preview_id` at `:501`. HC3 approves that ID and `apply` refuses any other at `:528`. `apply` recomputes the actions itself but does not re-check that they still produce the approved ID; this is limitation L2 in Section 8.2. Figure 5-1 shows the complete write path with its refusals.

5.3 Read-back reconciliation

After every apply, `readback` reports missing, mismatched, duplicate, unlabelled and human-Done items, and sets `verified` only if there are no missing, mismatched or duplicate items, see `:568–592`.

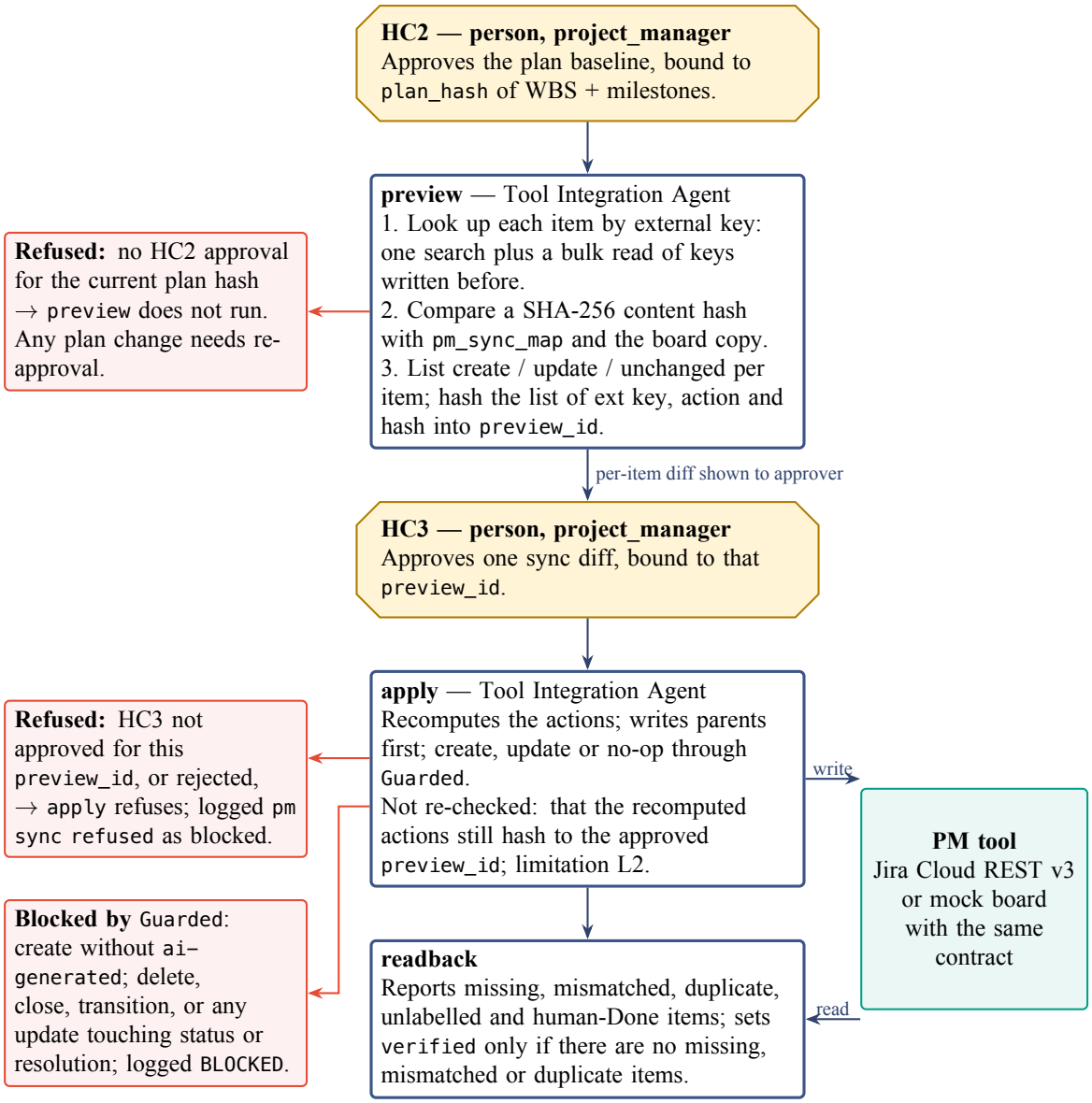


Figure 5-1 PM write path. HC2 must approve the current plan hash before `preview` runs; HC3 approves one `preview_id`; `apply` writes through `Guarded` and `readback` reconciles the board. Refusals are shown on the left; the code is in `app/pmtool.py:482–592`.

5.4 Allowed and refused operations

- **Allowed:** create; update of summary, description, labels, due date and parent; comment; link; and reads.
- **Refused and logged:** delete, close, transition, and any update touching status or resolution, see `:363–390`.
- **One exception:** the agent deletes an *issue link* it has just created in the wrong direction, and logs it, see `:308–312`. It never deletes an issue.

5.5 Reset demo: maintenance, not an agent operation

The start screen has a **Reset demo** button that is reachable before choosing an identity, see `server.py:408–415`. It is separate from the agents' `Guarded` adapter. Figure 5-2 summarises the four steps:

1. **Preview** through `GET /api/reset/preview` in `server.py:49–64`. It lists every issue in the configured Jira project, found by one project search plus the keys in `pm_sync_map`, so items missing from Jira's search index are included, see `app/pmtool.py:401–410`. The snapshot of site, project, Jira count, mock count and confirmation text is hashed into a reset `preview_id`.
2. **Confirm** through `POST /api/reset/restore` in `server.py:67–81`. The user must type the Jira project key, or `RESET` when only the mock board is configured. The server refuses a missing or wrong confirmation with 400, a second reset already running with 409 and `RESET_LOCK`, and a snapshot whose `preview_id` has changed since the preview with 409.
3. **Delete**, implemented in `app/pmtool.py:413–448`. The issue list is fetched again and must match the preview exactly. Issues are deleted deepest-first with `deleteSubtasks=false`, so nothing is removed by cascade, and each deletion is written to `audit_log`. The first Jira error stops the reset; the local data is then **not** reset and the error reports how many issues were already deleted.
4. **Restore**, implemented in `app/pipeline.py:83–100`. Only after all deletions succeed is every demo table cleared, including approvals, decisions, sessions and custom accounts; the four default accounts are seeded again; and a baseline run without the LLM rebuilds the outputs. `audit_log` is kept.

This deletes **every** issue in the project, including issues the system did not create, so it must only be pointed at the team's sandbox project; this is limitation L8 in Section 8.2. An offline check on the mock board on 14 Nov 2026 confirmed the refusals and that the audit log survives, see `docs/governance/evidence/reset_check.json`. The Overview page keeps the older, admin-only **Reset demo outputs** in `app/pipeline.py:72–80`, which clears outputs but keeps approvals and decisions.

5.6 Jira endpoints, the Jira run and read before write

Jira endpoints. The agent uses ten Jira Cloud REST v3 endpoints for search, bulk fetch, issue-type metadata, issue create and update, comment, issue links, the current user, project and link types; they are implemented in `:209–341`. Reset demo also deletes issues through the Jira API;

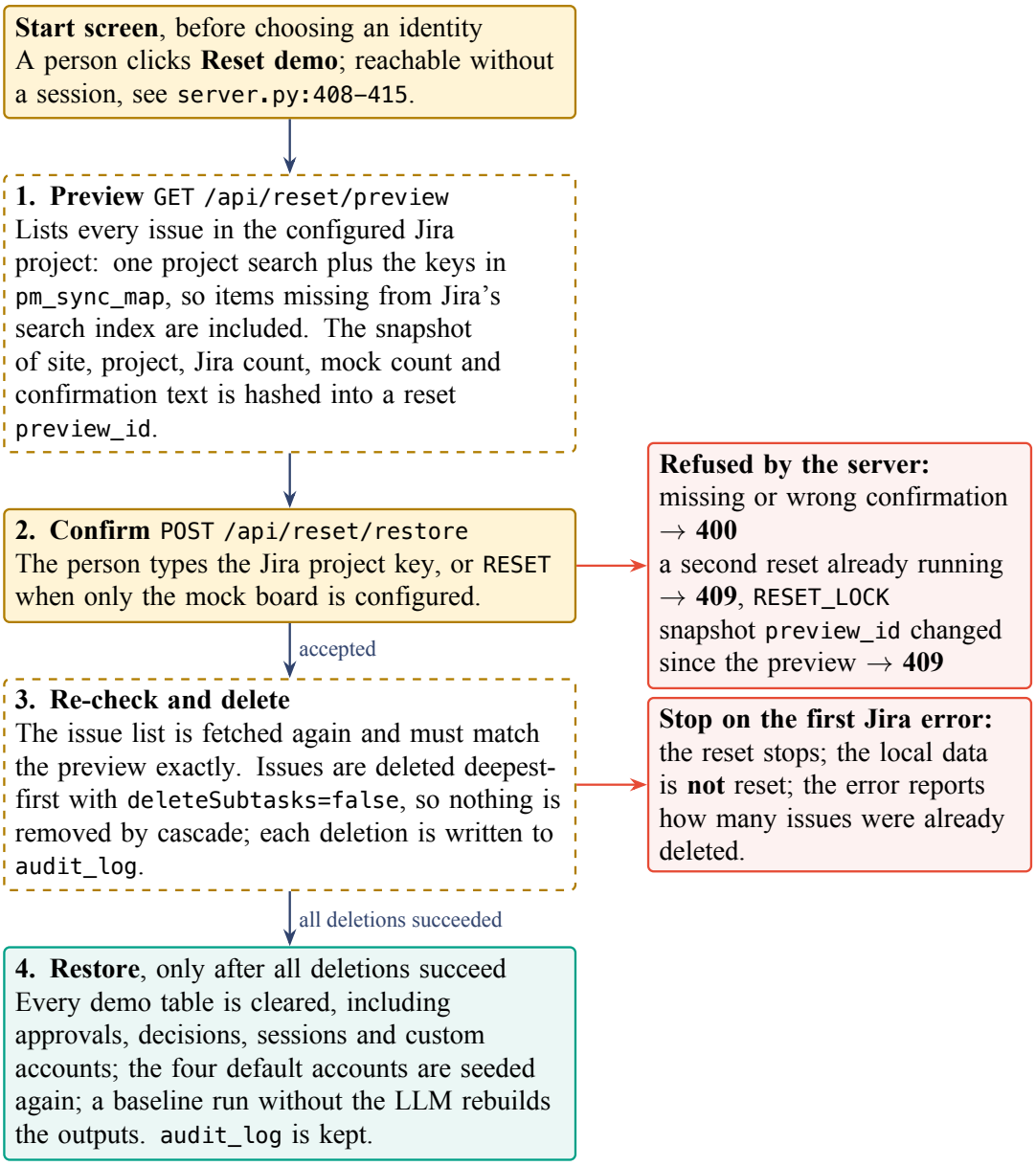


Figure 5-2 Reset demo flow, a human maintenance action outside the agents' `Guarded` adapter. Refusals and the stop rule are shown on the right; `audit_log` is never cleared.

the agent never does. Rate limits 429 and 503 are retried using Retry-After at `:162–163`, a standard stability pattern for calls to an external service (Nygard, 2018).

Jira run. The team's own Jira Cloud site on the Free plan, with project key `SCRUM`, is the sandbox; the API token is Tian Yujia's. According to the team, it ran HC2/HC3-approved writes and read-backs from its demo machine, where the audit and approvals exports are kept. A read-only snapshot of the project taken from this repository on 14 Nov 2026, `docs/governance/evidence/jira_snapshot_2026-11-14.json`, shows the result of the latest sync: 92 issues `SCRUM-283`–`SCRUM-374`, created 16:51–16:52 SGT, all labelled `ai-generated`, and 92 distinct external keys with no duplicates. There are 6 Epics, 37 Tasks and 49 Subtasks on a site with Chinese type names 长篇故事 / 任务 / 子任务, 68 items with a parent, 18 milestones and 18 "Blocks" links, all in the initial status. This matches the baseline plan item for item. Keys start at 283, so earlier syncs had already been made and cleared with Reset demo.

Read before write. The read path always runs before a write, by construction: `apply` refuses anything but an HC3-approved `preview_id`, see `app/pmtool.py:528–531`, and `preview` reads the board first with one JQL search plus a bulk read, see `app/pmtool.py:482–501` and `:452–479`. `scripts/jira_check.py` and the PM tool page's **Test Jira connection**, which calls `health` in `app/pmtool.py:599–612`, are read-only checks of account, project, issue types and link types.

6 Change handling and staleness

The Change Agent, whose card is Table 3-7, handles a change event in five stages:

- **Provenance.** Every stored output carries `tags` from a controlled vocabulary, see `app/gen.py:14–16` and `app/db.py:1–2`. Generated items may only use these tags, see `app/gen.py:57–58`.
- **Detection.** A change event such as `CE–01` is injected as a dated, numbered source without editing the pack, see `app/change.py:236–245`. Its tags come from keyword rules in `app/change.py:21–43`.
- **Matching.** An item is stale when it shares a primary tag with the event, or two broad tags, see `app/change.py:251–255`. Milestones downstream of an affected milestone are found by a dependency walk in `app/change.py:51–58`. Stale risks, what-ifs, reports and milestones are written to `stale_items` with a reason, see `app/change.py:267–282`. Detected conflicts that match the event by the same rule are always listed as UNRESOLVED in the Change Impact Note, whether or not the generated analysis names them, see `app/change.py:285–290`.
- **Refresh without a manual re-run.** What-ifs and report drafts regenerate immediately; the new reports must cite the event or the Critic stops them under rule C5, see `app/change.py:295–306` and eval T25, T28.
- **Waiting for HC6.** Risk re-scores, milestone notes, assumption updates and PM comment + label changes are drafted as pending actions and applied only after HC6, see `app/change.py:348–431`. An approved re-score makes the live reports stale and regenerates them, see `app/change.py:399–405` and eval T26.

6.1 Week 12 run

The team's event is **CE-01**, because the last digit of the team lead's matriculation number, G2510505D, is odd. The team confirmed that the wording in `change_events/CE–01.json` matches the Week 12 Guide. According to a team statement of 14 Nov 2026, the team ran CE-01 on its demo machine, where the records are kept. The committed evidence is the evaluation run, eval T24–T28 in baseline mode, and the governance demonstration run in `docs/governance/evidence/change_impact_CE–01.md`.

7 Governance, responsible AI and free-tier compliance

This chapter covers the audit trail, the guardrails, token hygiene, the bias and fairness check and the tools used, which together are the system's responsible-AI controls (cf. Tabassi, 2023).

7.1 Audit log

The table `audit_log(id, ts, run_id, actor, actor_type agent|human|system, action, target_type, target_id, outcome ok|blocked|rejected|fallback|error, detail JSON)` is defined at `app/db.py:61-64` and written by `audit.log` in `app/audit.py:43-49`. Approvals and hand-offs have their own tables. Audit and approvals export as CSV, and each export is itself logged, see `server.py:432-442`. Nothing in the application deletes `audit_log` rows.

Reset demo outputs, an admin-only action on the Overview page, keeps approvals and the decision log, see `app/pipeline.py:72-80`. **Reset demo** on the start screen clears approvals and decisions as well but keeps `audit_log`, which already mirrors every approval, see `app/pipeline.py:83-100`. Export the approvals CSV before a reset if the table itself is needed. Eval T30 found 92 PM write rows for 92 issues and no unnamed approvals.

7.2 Guardrails

There is one project key, and agent search is scoped to the project + `ai-generated`, see `app/pmtool.py:319-320`. The label is enforced on every create at `:357-358`. Agents never move items to Done, close or delete them, as Chapter 4 sets out. Unknown API paths default to the most restrictive permission in `server.py:370-378`, except the start-screen calls `/api/identities`, `/api/reset/preview` and `/api/reset/restore`, which are answered before any session check in `server.py:408-415`. Generated IDs, scores, dates and numbers are validated before storage, as the agent cards in Chapter 3 show.

7.3 Token hygiene

Keys come only from the environment or a git-ignored `.env`, see `app/config.py:13-24`; `.env.example` holds no values. `audit.redact` masks token patterns before logging, see `app/audit.py:33-40`. The Jira health check returns `token_set` as a boolean only, see `app/pmtool.py:599-612`. LLM keys travel only in the request header, see `app/llm.py:84-91`. A history scan found no committed credential.

7.4 Bias and fairness check

The check is implemented in `app/governance.py:43-107`. The 214 consultation responses are theme-tagged in four languages; raw theme ranks are compared with age-re-weighted and English-only ranks, and channel, age and language skew is reported as findings B1–B6. Residents 65+ are under-represented, an English-only summary would drop most active-ageing

and lift comments, and re-weighting changes the top-four order, as eval T29 checks. Five actions follow, including holding the summary D15 for HC7.

What changed as a result: at HC7 the team approved and carried out the actions except the most critical one, the second consultation round for residents 55+, which was not approved; Governance Log §3 gives the detail.

7.5 Tools and free tiers

Table 7-1 lists each tool, its tier and its status under the brief.

Table 7-1 Tools, free tiers and brief status

Tool	Used for	Tier / cost	Brief status
Gemini via Google AI Studio, `gemini-2.5-flash`, configurable	Primary LLM for all generation	Free tier; price set to 0 in `app/config.py:37-38`	Permitted, L210–L212
DeepSeek `deepseek-chat`	Backup LLM, used only when Gemini fails	Pay-per-token; about US$0.026 per full fresh run, measured in the LLM-mode evaluation of 14 Nov 2026	Not on the brief's list, L210–L212; **use accepted by the course instructor** according to the team. It can be switched off with `HCH_LLM_ORDER=gemini`; every required feature also runs in baseline mode at 0 tokens, see `eval/results/latest.md`
Jira Cloud, the team's own site, project `SCRUM`	PM tool	Free plan	Recommended, L232
Python 3 stdlib, SQLite, vanilla JS	Code, storage, UI	Free	Permitted "with code or without it", L205
OpenAI Agent Builder / ChatGPT agent mode	Not used	—	Prohibited, L239–L240; no such dependency

The vendor-change risk is logged: R23 in the baseline, and enforced by the validator in generated registers, see `app/risk.py:104-106` and `228-234`. The team re-checked the Gemini, DeepSeek and Jira terms in Week 12 and, according to a team statement of 14 Nov 2026, found no change that affects the project.

8 Design trade-offs and limitations

8.1 Design trade-offs

Table 8-1 records each major design decision, the alternative that was rejected, and what the decision costs.

Table 8-1 Design trade-offs

#	Decision	Alternative rejected	Reason	Cost
T1	LLM generates structured JSON; deterministic validators and Critic decide; curated baseline as fallback	Free-text LLM output reviewed by people only	Every number becomes checkable; failures degrade to a known-good output, not a blank page	Generated plans differ between fresh databases, and only the per-DB cache makes re-runs identical; baseline tables must be maintained
T2	Plain Python stdlib orchestration	OpenAI Agents SDK, CrewAI or LangChain, permitted by L229–L230	No dependencies to install; hand-offs and gates are explicit code and logged	No framework tracing or tool-calling; the team writes its own loop, the Critic revision loop
T3	Mock board with the same contract as Jira	Jira-only development	Evaluation and demos run offline without touching the sandbox	Mock evidence is not Jira evidence, see Chapter 5
T4	Re-scores and change updates wait for HC6	Auto-apply re-scores	Scores drive the top-five list in external reports; the brief asks for actions drafted for a human	Stale scores persist until someone approves
T5	BM25 + rule-based sufficiency test for NOT FOUND	Embedding search + LLM judgement	Deterministic, explainable, works with no LLM	Brittle on paraphrase; one case found and fixed, L3
T6	Tag-based staleness + milestone dependency walk	Full provenance graph per claim	Simple and auditable; tags are validated against one vocabulary	Coarse: may over-flag; WBS rows are flagged, not regenerated
T7	Passwordless identity selection and a one-click project reset on the start screen, added 11 Nov 2026	Password login, commit `8044577`; manual clean-up in Jira between demo runs	Faster demonstrations: switching role to show RBAC takes one click, and a demo can be repeated from a clean board	Approvals no longer prove who approved, L7; anyone who can open the page can empty the sandbox project, L8

8.2 Known limitations

L1: Generated Sponsor Questions recall. Kept. In LLM mode the generated register recalls 77–82% of the team's 22 gold gaps across runs, and 82% in the final run; baseline mode recalls 91%. The team's own target is 80%; the course instructor later confirmed a lower threshold, which every run meets, according to the team. The evaluation checks that once pinned baseline

IDs were rewritten as behaviour checks, and LLM mode now passes 34 of 34; the logs are in `docs/evaluation/evidence/`.

L2: HC3 diff binding. Accepted by the team. `apply` does not verify that the current diff still hashes to the approved `preview_id`, so a plan changed after HC3 but re-approved at HC2 could be written under an older HC3; see `app/pmtool.py:526–565`.

L3: Memory false NOT FOUND. Fixed 14 Nov 2026. "How often does the EDC meet?" used to return NOT FOUND although `EDC-1.6` and `EDC-2.3` answer it, because "often" was counted as a key term. Frequency questions are now recognised in `app/memory.py` and must be answered by a sentence that states a frequency such as weekly or fortnightly. The answer cites both sources; eval T34 checks it.

L4: Checkpoints that only record. Accepted by the team. HC1, HC4 and HC7 are recorded and shown but gate no code path, as Chapter 4 shows.

L5: Evidence not in the repository. The Jira approvals and the team's Week 12 run are recorded on the team's demo machine, not in the repository; the repository holds a read-only Jira snapshot. Gemini has not been run; all LLM evidence uses DeepSeek.

L6: Partial stretch items. The three what-ifs are shown as cards, not a comparison table. Re-scoring is triggered by change events, not by arbitrary plan edits.

L7: Passwordless demo identities. Anyone who can reach the app can select an active account, including admin. Role permissions still apply to the selected session, but approvals are not proof of a person's identity. This reopens governance incident I-04, described in Governance Log §5. Use a protected local demo environment, and restore authentication, which is still in history at commit `8044577`, before using this workflow outside the team. **Team decision of 14 Nov 2026: accepted**, because the team alone controls the demo machine; approvals are attributed to role accounts, not named people.

L8: Project-wide reset. Anyone who can reach the identity screen can preview and confirm deletion of every issue visible to the configured Jira account in the selected project. The operation is irreversible and may partially complete if Jira refuses a later deletion. The local baseline is restored only after all requested Jira deletions succeed. The reset also clears the `approvals` and `decisions` tables; `audit_log` keeps the history. The Jira-side deletion has not been run in this repository's evidence; only the mock-board path was checked. **Team decision of 14 Nov 2026: accepted**, because `SCRUM` is the team's own sandbox and holds only system items; approvals are exported before a reset.

References

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Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK guide) and the standard for project management* (7th ed.).

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A Assumption register

Seeding. There is no standalone assumption file. The register is the Sponsor Question table:

- Every question not answered by a source gets `ASSUMPTION A-nn: until the sponsor answers, plan treats this as unresolved…` see `app/planning.py:513`.
- Approved HC6 actions append event notes, for example "(CE-01)", see `app/change.py:420-427`.
- The rows below are the **baseline** register GQ01–GQ20 from `app/planning.py:45-181`. In LLM mode the IDs and wording are regenerated on each fresh database; the live register is on the Planning page.
- Rows marked **added** cover G01–G14 items that the baseline register does not hold as a question.

Owners are role accounts; owners and revisit triggers were adopted by the team lead.

Table A-1 Baseline assumption register

ID	Assumption or open question	Sources	Sponsor answer	Impact	Owner	Revisit trigger
A-01	Fixed completion date: 22 months from possession vs fully open by Q4 2028 vs KHB 2028-12-31 vs D22 2028-10-31	BRIEF-p3-5, EDC-4.1, DEL-D22, NRF-4.2	SA-Q1 confirms possession from 4 Jan 2027; no single completion date given — remains assumption	Schedule; NRF approval needed after 31 Dec 2028	`project_manager`	Programme revision or NRF 4.2 request
A-02	Grant is 60% of eligible cost capped at S$17.0M, max S$10.2M, not 60% of S$18.5M	BRIEF-p3-5, NRF-1.1, NRF-1.2, NRF-1.3	Unanswered — remains assumption	Council share above 40% of the budget; derived: 18.5 − 10.2 = S$8.3M	Finance	Fund eligibility ruling, e.g. CE-01
A-03	∼S$1.18M headroom is not yet contingency; drawdown authority undecided	EDC-1.1, EDC-1.2	Unanswered — remains assumption	Asbestos S$0.35–0.60M would use 30–51% of it	Finance	EDC decision on contingency
A-04	Childcare handover needed about 7 months before the Jan 2029 intake vs programme month 20	BRIEF-p3-6, EDC-4.1, DEL-D21, LL-L04	Unanswered — remains assumption	Childcare opening and T3/T4 timing	`project_manager`	Operator licensing plan
A-05	No closures are agreed; east-core closure would break barrier-free access to L3–4	BRIEF-p3-6, EDC-3.1, NRF-5.1	SA-Q3: no closures agreed; priority is the sponsor's	6–8 weeks delay if lift works slip, PR3-3	Sponsor	Closure decision at EDC
A-06	Active ageing operator appointed by Council; EOI needs 3 responses, has 1	EDC-4.4, DEL-D14	SA-Q2: Council appoints; no confirmed date	Tranche 2 needs ALL operator agreements, NRF-T2	Community Engagement	EOI close 2027-04-30 or CE-02
A-07	Asbestos removal eligibility unknown	EDC-5.1, EDC-6.5, PR3-4	Unanswered — remains assumption; CE-01 rules it ineligible if that event applies	Council exposure 40–100% of S$0.35–0.60M, W1	Finance	Fund ruling
A-08	Rooftop garden kept or deferred; garden ineligible; slab strengthening unpriced	NRF-1.3, EDC-5.2, EDC-6.3	Unanswered — remains assumption	Cost NOT FOUND; W3	Sponsor	EDC vote

Table A-1 continued

ID	Assumption or open question	Sources	Sponsor answer	Impact	Owner	Revisit trigger
A-09, added	EDC cadence: weekly in the brief vs fortnightly from February with alternate-week written updates	BRIEF-p3-7, EDC-1.6, EDC-2.3	Unanswered — remains assumption	Reporting calendar for the EDC update	`project_manager`	Next EDC terms of reference
A-10, added	Monthly NRF report due by the 10th working day; February report was late; a second late report risks suspension	NRF-3.1, NRF-3.3, DEL-D09	Not a sponsor question; data pack rule	Tranche suspension	`group_leader`	Each monthly deadline
A-11, added	KHB “broadly on programme” is treated as a claim, not a status; figures show −2.5% physical, −9% design	PR3-1, PR3-2	n/a	Reports must attribute the wording to KHB, Critic C6	`group_leader`	Each contractor report
A-12, added	Seed register defects are repaired, not inherited: R01 owner left, R03/R04 duplicate, R06 score error, R08 closed but realised, R11 no owner	RISK-R01, RISK-R03, RISK-R04, RISK-R06, RISK-R08, RISK-R11	n/a	Register accuracy	`group_leader`	HC4
A-13, added	Deliverables schedule data defects are flagged, not corrected: mixed date formats, D06 no due date, D20 no actual date, D14 no owner	DEL-D03, DEL-D06, DEL-D10, DEL-D14, DEL-D20	n/a	Milestone dates; T2 owner	`project_manager`	Next schedule issue from KHB
A-14, added	Values the pack does not contain stay NOT FOUND: Finance Manager funding figures, temporary lift price, roof strengthening cost, garden cost, energy target	EDC-3.5, EDC-3.2, EDC-5.2, BRIEF-p3-3	Unanswered — remains assumption	What-ifs W2/W3 show NOT FOUND	Finance	Sponsor provides figures

Full baseline GQ01–GQ20 register, with status, sponsor answer and assumption: `docs/design/evidence/sponsor_questions_register_baseline.csv`. Its `gaps_hash` is `072aec9495dc`, the hash approved at HC1 in the demonstration run. Only Round 1 sponsor answers exist in the data pack, in `data_pack/00_Sponsor_Answers_Round1.txt`.

B Agent prompts: Role and Context as stored

The source is `app/agents.py`. No tokens appear in any prompt.

Shared grounding text, in `app/agents.py:4–7`: "You may ONLY use the SOURCES given in the user message. Never use outside knowledge. If the sources do not contain the answer, say exactly NOT FOUND. Every claim must carry a citation in square brackets using the source id, e.g. [EDC-5.1]. When sources conflict, say so and cite both."

Generation grounding, in `app/gen.py:73–76`: "Use ONLY the SOURCES given. Never use outside knowledge or invent names, dates, amounts or owners. Every source id you output must be one of the ids shown in square brackets. If the sources do not state something, write NOT FOUND […] instead of guessing. Where sources disagree, say so and cite both. Return a single JSON object only."

Table B-1 gives the Role, Context and system prompt of each agent.

Table B-1 Agent prompts: Role, Context and system prompt

Agent	Role	Context	System prompt
Planning	Project planner for the Harbourline Community Hub Refurbishment.	Brief + Sponsor Answers Round 1 + data pack only. Relative time until the sponsor gives dates.	"You are the Planning Agent. " + grounding; generation: "You are the Planning Agent for the Harbourline Community Hub Refurbishment. " + generation grounding, `app/planning.py:323–324`
Tool Integration	Writes the approved plan into the PM tool and reads it back.	Approved WBS/milestones only, from HC2. One sandbox project. Label ai-generated. Stable external keys.	"Deterministic agent - no LLM."
Memory & Retrieval	Answers questions over the pack with document + line citations.	BM25 retrieval over chunks; human-approved decisions are carried forward between sessions.	"You are the Memory Agent for the Harbourline project. " + grounding + " Answer in at most 4 sentences."
Risk & What-if	Maintains the scored risk register and runs what-if scenarios.	Seed register of 2025 + pack evidence + WBS handed off from Planning.	"You are the Risk & What-if Agent. " + grounding; generation prompt `app/risk.py:180–202`
Reporting	Drafts the weekly EDC exception update and the monthly NRF progress report from ONE fact sheet.	Fact store + risk register + tranche conditions. Audience-fit: EDC = one-page exception report, L07; NRF = clause 3.2 contents.	Summary: "You are the Reporting Agent. Rewrite the given bullet facts into a short executive summary. Do not add any number, date, name or claim that is not in the bullets. Keep the [F..] fact tags." Generation prompt with drafting rules: `app/reporting.py:167–192`
Critic	Checks every draft before a human sees it.	Fact store, risk register, required report contents, stale-fact list.	"Deterministic agent - no LLM."
Change	Responds to the Week 12 change event.	Event text, dated and numbered, + every stored output's tags.	"You are the Change Agent. " + grounding; analysis prompt `app/change.py:130–158`

Table B-1 continued

Agent	Role	Context	System prompt
Governance	Audit log, checkpoints, guardrails, bias and fairness check.	All agent writes; consultation CSV.	“Deterministic.”

The UI translation helper in `app/translation.py` has its own display-only prompt and never feeds an agent.

C Coverage table, updated 14 November 2026

Status: **Implemented** = code + run evidence in the repo; **Partial** = code exists but part of the requirement or its evidence is missing. Table C-1 maps each requirement of the brief to its evidence; Figure C-1 summarises the status counts.

Table C-1 Coverage of the brief's requirements

#	Requirement	Brief ref	Status	Evidence	Note
1.1	Scope statement + Sponsor Questions	L137-143	Implemented	`app/planning.py: 327–377`, `574–584`; T01	LLM-mode recall 77–82%, L1
1.2	WBS to L3 + dictionary	L137-143	Implemented	`app/planning.py: 380–420`, `564–571`; T04: baseline 49 L3; LLM 61 L3 in the final run	
1.3	Milestones with dependencies	L137-143	Implemented	`app/planning.py: 435–474`, `595–605`; T05	
1.S	Traceable WBS; untraceable flagged	L137-140	Implemented	`app/planning.py: 518–536`; T03	
2.1	Write WBS + milestones behind confirmation	L145-150	Implemented	`app/pmtool.py:482–565`; T21, T22 on mock; Jira snapshot of `SCRUM`	92 items on Jira; approval exports on the team machine
2.2	Board read back	L149-150	Implemented	`app/pmtool.py:568–592`; T22	Jira read-back run by the team, team statement
2.S	Stable external keys	L145-148	Implemented	`app/pmtool.py:37–38`, `452–479`; T22	
2.x	Approval bound to the diff	writer brief §5	Partial	`app/pmtool.py:501`, `528`	L2
3.1	Answers with document + line citation	L152-155	Implemented	`app/memory.py:100–139`, `app/sources.py:249–255`; T06–T09	
3.S1	NOT FOUND when silent	L152-154	Implemented	`app/memory.py:76–87`; T10–T13, T34	L3 fixed
3.S2	Decisions carried across sessions	L154-155	Implemented	`app/memory.py:142–154`, `app/pipeline.py:13`; T15	
4.1	Scored register, owners, triggers, linked to work	L157-163	Implemented	`app/risk.py:173–279`; T16	
4.2	One what-if with cost + schedule	L160-163	Implemented	`app/risk.py:301–381`; T17	
4.S1	Three what-ifs compared	L157-158	Partial	W1–W3 `app/risk.py:311–357`	No comparison output
4.S2	Automatic re-scoring	L158-159	Partial	`app/change.py:150–151`, `app/risk.py:394–419`; T26	Event-driven; applied after HC6
5.1	Two reports from the same facts	L177-183	Implemented	`app/reporting.py: 161–288`; T18	

Table C-1 continued

#	Requirement	Brief ref	Status	Evidence	Note
5.2	Critic before a human; stop condition	L180-183	Implemented	`app/critic.py:123–169`, `app/reporting.py: 221–238`, `291–298`; T19, T20	Revision loop max 3
5.S1	Critic findings logged	L177	Implemented	`app/reporting.py: 231–233`, `285–287`	
5.S2	A report the Critic stopped	L178-179	Implemented	`eval/faults.py`; T19, T28	Planted faults
6.1	Detect the Week 12 change	L185-190	Implemented	`app/change.py:236–245`; T24, T27	Team's event is CE-01; team run on the demo machine, team statement
6.2	Show what is stale	L186-189	Implemented	`app/change.py:248–282`; T24	
6.3	Draft actions for a human	L189-190	Implemented	`app/change.py:348–431`; T26	
6.S	Stale outputs refresh without re-run	L185-187	Implemented, partial scope	`app/change.py:295–306`; T25	WBS/milestones via HC6
G.1	Named human checkpoints	L192-196	Implemented	`app/audit.py:16–31`, `server.py:118–143`	HC1/HC4/HC7 record only, L4; approver is an account chosen without a password, L7
G.2	Audit log of every agent write	L193-194	Implemented	`app/audit.py:43–49`; T30	Planning logs one summary row per run
G.3	Never delete/close/Done	L194-196	Implemented	`app/pmtool.py:343–390`; T23	Agents only. Link-direction fix disclosed; the human Reset demo deletes all project issues, L8
G.S	Bias and fairness check	L192-194	Implemented, acted on	`app/governance.py: 43–112`; T29	HC7: all actions but the second consultation round
I.1	Hand-off without human copying	L199-200	Implemented	`app/audit.py:52–57`; T31	
I.2	Every approval point designed in	L200-201	Implemented	Chapter 4	L4
R.1	Free tiers only	L206-207	Implemented, approved exception	`app/config.py:27–59`; baseline 0 tokens	Jira Free; DeepSeek is paid, accepted by the course instructor
R.2	One sandbox project	L245	Implemented	`app/config.py:72`, `app/pmtool.py:319–320`	Project `SCRUM` on the team's own site
R.3	`ai–generated` label	L245-246	Implemented	`app/pmtool.py:357–358`, `585`	
R.4	Personal tokens	L249-250	Implemented	`app/audit.py:33–40`, `.gitignore`; history scan clean	Jira token and LLM keys are Tian Yujia's
R.5	Read path before write access	L251	Implemented	`app/pmtool.py:482–501`, `528–531`, `599–612`, `scripts/jira_check.py`	Enforced: every write needs a preview, which reads first; see Chapter 5

Table C-1 continued

#	Requirement	Brief ref	Status	Evidence	Note
R.6	No Agent Builder / agent mode	L239-240	Implemented	stdlib only	
R.7	Vendor-change risk logged	L240-241	Implemented	`app/risk.py:104–106`, `228–234`	Week 12 re-check done, no change, team statement
R.8	Code orchestration tool	L229-230	Note	Plain Python	Trade-off T2

Counts, 37 rows: 33 Implemented, including the "partial scope", "acted on" and "approved exception" variants; 3 Partial: 2.x, 4.S1 and 4.S2; 1 note. Planned 0, Missing 0. Since the 8 Nov version, 2.1, 2.2, 2.S, R.1, R.2 and R.5 moved to Implemented on the evidence of the Jira snapshot, the code-enforced read path and the team's statements of 14 Nov 2026.

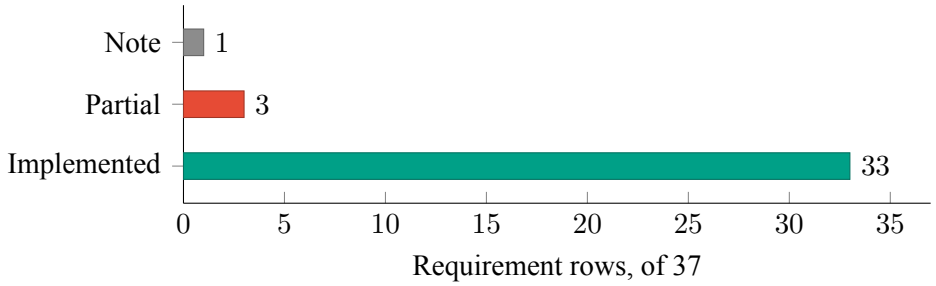


Figure C-1 Coverage status counts from Table C-1; "Implemented" includes the partial-scope, acted-on and approved-exception variants

D AI-use declaration

Table D-1 declares, for each deliverable, which AI tools were used, what they produced and how people reviewed it.

Table D-1 AI-use declaration

Deliverable	AI tools and what they produced	Human review
Design document: this file, `architecture.mmd`, inventory, claims trace, open items	Claude by Anthropic, used through Claude Code with model Opus 5.5, drafted the text, tables and diagram from the repository and data pack on 8 Nov 2026, and updated them on 14 Nov 2026	Tian Yujia, the owner, reviewed the content against the system
System code: `app/`, `server.py`, `static/`, `scripts/`	Claude, through Claude Code, wrote code including LLM generation and the Gemini/DeepSeek provider chain; the passwordless identities and Reset demo by Tian Yujia with Claude	Tian Yujia, demo system owner
Agent prompts: `app/agents.py`, inline generation prompts	Claude, through Claude Code	Tian Yujia
Runtime outputs: plan, register, reports, change analysis	DeepSeek at run time, with Gemini configured as primary but not used, validated by deterministic code	HC1–HC7 approvals by the role accounts, see Chapter 4
Evaluation report	Updated by Claude, through Claude Code, on 14 Nov 2026 after re-running the harness; owner Qu Chengzhi	Qu Chengzhi
Governance log	Updated by Claude, through Claude Code, on 14 Nov 2026 after regenerating the demonstration evidence; owner Huang Hao	Huang Hao
Slides	Liu Beiqi; ChatGPT used to check the slides for errors	Liu Beiqi
Demo video	Zhang Yinan	Zhang Yinan

E Source-ID and WBS-code conventions

Table E-1 lists the identifier patterns used throughout the system and this document.

Table E-1 Source-ID and code conventions

ID pattern	Meaning	Example	Defined in
BRIEF-p<page>-<para>	Brief paragraph	BRIEF-p3-5	app/sources.py:53-77
SA-Q<n>	Sponsor Answers Round 1	SA-Q3	app/sources.py:80-102
EDC-<item>	EDC minutes item	EDC-2.3	app/sources.py:105-133
PR1-/PR2-/PR3-<§>	Contractor report Jan/Feb/Mar 2027	PR3-3	app/sources.py:136-151
DEL-D<nn>	Deliverables schedule row	DEL-D22	app/sources.py:194-210
RISK-R<nn>	Seed risk register row	RISK-R08	as above
RC-RC<nnn>	Consultation response	RC-RC002	as above
LL-L<nn>	Seabreeze lesson	LL-L04	app/sources.py:154-165
NRF-<clause> / NRF-T<n>	Grant schedule clause / tranche	NRF-4.2, NRF-T2	app/sources.py:168-191
F<nn>	Verified fact with exact quote	F24	app/facts_catalog.py
CE-<nn> / DL-<nnn>	Change event / human decision	CE-01, DL-001	app/change.py:236-245, app/memory.py:142-149
CF<nn>	Conflict register	CF12	app/planning.py: 219-281
SC<nn> / OS<nn>	Scope line in / out of scope	SC03	app/planning.py: 330-331
GQ<nn> / A-<nn>	Sponsor Question / its assumption	GQ09 → A-09	app/planning.py:513
MS<nn>	Milestone; M+n = months from site possession	MS07, M+22	app/planning.py:440
R<nn>	Live risk; R01–R14 keep seed IDs, R15+ are new	R22	app/risk.py:185
W<n>	What-if scenario	W1	app/risk.py:301-357
HC<n>	Human checkpoint	HC3	app/audit.py:16-31
B<n>	Bias finding	B3	app/governance.py: 66-91
ext-HCH-WBS-<code> / ext-HCH-MS<nn>	PM external key, stored as a label	ext-HCH-WBS-1 -2-3	app/pmtool.py:37-38

WBS codes: 1 is Level 1, an Epic → 1.2 is Level 2, a Task → 1.2.3 is Level 3, a Subtask; there is no Level 4, see app/planning.py:386. Owners UNASSIGNED (<why>) are kept visible, not filled in.